

Database Management System (DBMS) Architecture

Q. Explain the different types of Database Management System (DBMS) Architectures. Differentiate between Two-Tier and Three-Tier architecture.

[DEF] Definition to Remember

DBMS Architecture is the fundamental design structure of a database system that determines how data is stored, processed, and accessed by users and applications. It is classified into One-Tier, Two-Tier, and Three-Tier architectures based on how the application connects to the database.

1. Components of DBMS

A typical DBMS architecture consists of:

2. Types of DBMS Architecture

DBMS architectures are classified into three tiers:

1. One-Tier Architecture

2. Two-Tier Architecture (Client-Server)

3. Three-Tier Architecture

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[Client Tier]
Web Browser / App
    |
[Application Layer]
Business Logic Server
    |
[Database Layer]
DBMS + Physical DB
  
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3. Difference: Two-Tier vs Three-Tier

Feature	Two-Tier Architecture	Three-Tier Architecture
Layers	Client + Database Server	Client + Application Server + Database
Application Logic	On the Client machine	On the middle Application Server
Security	Lower (direct DB access)	Higher (DB hidden behind app server)
Scalability	Poor	Excellent

Best Suited For	Small LAN applications	Web/enterprise applications
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[MUST-WRITE] Must-Write Points (for 10 marks)

1. DBMS Architecture defines how users, applications, and the database interact.
2. One-Tier: single machine; Two-Tier: client-server; Three-Tier: client + app server + DB server.
3. Two-Tier uses ODBC/JDBC for direct client-to-database connection -- called a "fat client."
4. Three-Tier's middle Application Layer enforces business logic and acts as a security buffer.
5. Three-Tier is more secure because the client never directly accesses the database.
6. Two-Tier suits small networks; Three-Tier suits large-scale web applications.
7. Three-Tier is the modern industry standard for enterprise and web-based systems.

[RECALL] Quick Recall

One-Tier (1 machine) -> Two-Tier (Client + DB Server, ODBC/JDBC) -> Three-Tier (Client + App Server + DB, scalable & secure)